

Adaptive Bayesian Very Short-Term Wind Power Forecasting Based on the Generalised Logit Transformation

Reproducibility Package

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Package Overview

This package is provided for the reproducibility check of the manuscript submitted to the *International Journal of Forecasting (IJF)*.

The package contains the following items:

- [README.html](#) and [README.pdf](#): this document in HTML and PDF formats.
- [./src/](#): the source code used to reproduce the results.
- [./data/](#): the raw data used in the study.
- [./reference/](#): the reference intermediate outputs, figures, and tables generated by the authors.
- [IJF-D-25-00276R2_Manuscript.pdf](#): the manuscript (peer review) version, including all appendices, in PDF format.
- [COPYRIGHT](#): copyright information and permitted use of this package solely for reproducibility assessment.

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Data

Data Availability Statement

The data used in this paper were obtained from the Balancing Mechanism Reporting Service (BMRS) API provided by Elexon® (replaced by Insight Solution® in May 2024).

The dataset consists of electricity production records from 128 wind farms in Great Britain from 2020 to 2023 inclusive, with a time resolution of 30 minutes.

The raw data used in this study are included in this reproducibility package and can be found in the `./data/` directory.

Data Files

The raw data files included in this reproducibility package are listed below.

Data File	Source	Provided
<code>./data/UBOR_2020.csv</code>	BMRS API (Elexon®)	Yes
<code>./data/UBOR_2021.csv</code>	BMRS API (Elexon®)	Yes
<code>./data/UBOR_2022.csv</code>	BMRS API (Elexon®)	Yes
<code>./data/UBOR_2023.csv</code>	BMRS API (Elexon®)	Yes

Data Description

A detailed description of the dataset and its statistical properties is provided in Sections 5.1 and 5.2 of the attached manuscript.

Computational Environment

Hardware

The hardware environment used to generate the results is summarised in the table below.

Component	Specification
Operating System	Microsoft Windows 11 Pro (Build 26200)
System Manufacturer	ASUSTeK COMPUTER INC.
System Model	ROG Flow Z13 GZ301VU
Processor	13th Gen Intel® Core™ i9-13900H
Memory	16 GB RAM
Storage	2 TB NVMe SSD

The following observations are based on our limited experience with the computational environment used in this study and are provided for reference only.

- A machine with at least **16 GB of RAM** is recommended. We have not tested the reproducibility code on systems with less memory.
- For disk storage, approximately **20-30 GB of free disk space** should be sufficient to run the reproducibility package and store all intermediate files and reproduced outputs.
- High-performance system power profiles may reduce execution time.
- In our experiments, the main model evaluation script was executed using **4 logical CPU cores**, while the sensitivity analysis script was executed using **8 logical CPU cores**. Consequently, the reproducibility package requires a system with at least **8 logical CPU cores**.

The reproducibility code was executed on a **Windows 11 Pro** Operating System.

The reproducibility code was executed on a **CPU**.

The reproducibility code requires **parallel processing**. Therefore, a **multi-core CPU with at least 8 logical cores** is required. Systems with fewer than 8 logical cores are not supported.

The reproducibility code requires at least **8 GB of available RAM** at runtime. The reproducibility package will terminate if less available memory is detected.

Software

Python

The reproducibility package was developed using Python 3.12.0, but any Python **Python 3.12.x** version should be OK.

Required Python packages are

```
matplotlib==3.8.0
numpy==1.26.4
pandas==2.1.1
psutil==5.9.6
python-ternary==1.0.8
scipy==1.13.0
seaborn==0.13.0
statsmodels==0.14.5
ternary==0.1
```

.

The required Python packages can be installed using

```
pip install -r requirements.txt
```

.

Detailed instructions for reproducing the results are provided in the Reproduction Instructions section ([Reproduction Instructions](#)).

VS Code (Optional)

Visual Studio Code 1.121.0 was used as the integrated development environment (IDE) for code development and inspection. The use of VS Code is optional and is not required for reproducing the results.

Virtual Environment (Optional)

A virtual environment was not used during the development of this reproducibility package. However, replicators may choose to create and use one if preferred.

Total Runtime Estimation

The total runtime depends on the available hardware resources and system configuration. As a reference, the full reproduction process takes approximately **20 hours** on the system described above, using 4–8 CPU cores.

Detailed execution steps and options for running individual components separately are provided in the in the Reproduction Instructions section ([Reproduction Instructions](#)) section.

Code Description

Scripts List

The executable and auxiliary scripts contained in `./src/` are listed exhaustively below.

Relative Path	Description
<code>./RunAll.ps1</code>	PowerShell script that executes the complete reproduction workflow in the recommended order.
<code>./Scripts/_bootstrap.py</code>	Auxiliary bootstrap module containing shared initialisation utilities used by other scripts. It is not intended to be executed directly.
<code>./Scripts/RELEASE.ConfigCheck.py</code>	Verifies the computational environment, dependencies, directory structure, and required input files before reproduction, and creates the required output directories.
<code>./Scripts/RELEASE.DataQuality.py</code>	Performs data quality checks and generates summary statistics for the input datasets.
<code>./Scripts/RELEASE.RunModels.py</code>	Executes all forecasting models and generates the primary experimental results.
<code>./Scripts/RELEASE.RunSensitivity.py</code>	Performs the sensitivity analyses reported in the manuscript.
<code>./Scripts/RELEASE.FigureTable.py</code>	Generates all figures and tables reported in the manuscript from the reproduced results.
<code>./Scripts/RELEASE.FigureTable.ipynb</code>	Development notebook corresponding to <code>RELEASE.FigureTable.py</code> , provided for reference only.

The script `RunAll.ps1` is an all-in-one wrapper (referred to as the "Masterfile" in the guidelines). It sequentially executes all other scripts in the recommended order.

Detailed execution steps and options for running individual components separately are provided in the in the Reproduction Instructions section ([Reproduction Instructions](#)) section.

Expected Discrepancies

Rounding-Related Discrepancies in CRPS Tables

Minor discrepancies may occur in the CRPS summary tables due to differences in computational environments and the manual rounding process used when preparing the final manuscript tables.

The following files are affected:

- `Table.CRPS_Skill_Appendix.csv`

- [Table.CRPS_Skill_Mainpaper.csv](#)

The values reported in the manuscript were manually rounded and entered with a precision of three decimal places. Therefore, discrepancies of up to approximately $\pm 0.002\%$ should be considered acceptable.

For the rank table [Table.Rank.csv](#), no discrepancies are expected in principle. However, due to differences in floating-point arithmetic and computational environments, a discrepancy of ± 1 in individual rank counts should be considered acceptable.

Randomness-Related Discrepancies in Sensitivity Analysis

The sensitivity analysis was performed using parallel computation with **8 CPU cores** and random numbers generated by NumPy. Since fixed random seeds do not necessarily guarantee identical results across different hardware and software environments in parallel computing, no fixed random seed was enforced. As a result, the reproduced sensitivity-analysis results are expected to differ from the reference results.

Nevertheless, in [Fig.Sensitivity.jpeg](#), the following qualitative behaviours should remain consistent:

- For the disturbances on μ , σ_z^2 , and ν , the reproduced results should generally fluctuate around the blue reference values.
- For the disturbance on M , the results of **RLS**, **sRLS**, and **RMLE** should generally remain on the left-hand side (lower side) of the blue reference values.
- For the disturbance on M , the results of **sRMLE**, **Bayes**, and **Bayes- ν** should generally remain around the blue reference values.
 - For **Bayes** and **Bayes- ν** , the scale of the x-axis should remain smaller than 10^{-5} .

Reproduction Instructions

In this section, `./` denotes the root directory of the reproducibility package. When executing scripts, ensure that the current working directory in the terminal is the root directory of the reproducibility package.

Setup

Before running the reproduction scripts, please complete the following steps:

1. Prepare a suitable machine, virtual machine, or computational environment.
2. Install Python with recommended version.
3. Extract the reproducibility package. We recommend using a directory path that:
 - contains only letters (**a-z**, **A-Z**), numeric characters (**0-9**), and underscores (**_**);
 - contains **NO** spaces; and
 - has a total path length of fewer than **60** characters.
4. Install the required Python packages. The replicator may install them using

```
pip install -r ./src/requirements.txt
```

Configuration Customisation

The only parameter intended to be modified by replicators is `CPU_COUNT` in `./src/Configuration.py`.

Replicators may increase this value according to the number of available CPU cores. This parameter primarily affects the execution time. Minor numerical differences may occur due to differences in parallel execution environments, but no material impact on the reproduced conclusions is expected.

`RELEASE.RunSensitivity` uses a fixed setting of **8 CPU cores**. Replicators who wish to modify this value may edit the following line in `./src/Scripts/RELEASE.RunSensitivity.py` (line 389):

```
with Pool(processes=8) as pool: results = pool.map(expm, runs)
```

When increasing the number of CPU cores, we recommend doing so in pairs and providing at least **4 GB of available RAM for every 2 CPU cores**.

Running the Experiments

1. In PowerShell, run:

```
.\RunAll.ps1
```

The `RunAll` script automatically invokes the following scripts in the order:

- `RELEASE.ConfigCheck`
 - `RELEASE.DataQuality`
 - `RELEASE.RunModels`
 - `RELEASE.RunSensitivity`
 - `RELEASE.FigureTable`
2. `RELEASE.ConfigCheck` performs basic checks on paths, CPU cores, and available RAM. If an error is raised at this stage, please fix it accordingly. Alternatively, replicators may disable the checks at their own discretion. We do not recommend bypassing them.
 3. Log messages will be printed during execution. Replicators may choose to redirect them to a log file or leave them in the terminal.

4. During `RELEASE.RunModels`, it is expected to encounter the warning `ValueWarning: Matrix is singular. Using pinv..`
5. During `RELEASE.FigureTable`, it is expected to encounter approximately **1 UserWarning** and **8 ValueWarnings**.

Expected Runtime

The approximate runtime of each script on the reference system is summarised below.

Script	Expected Runtime
<code>RELEASE.ConfigCheck</code>	< 1 s
<code>RELEASE.DataQuality</code>	< 5 min
<code>RELEASE.RunModels</code>	~ 10 h
<code>RELEASE.RunSensitivity</code>	~ 6.5 h
<code>RELEASE.FigureTable</code>	< 30 min

The full reproduction workflow is expected to take approximately **20 hours** on the reference system.

Actual runtimes may vary depending on the hardware configuration, available memory, CPU core count, and system load.

Results

In this section, `./` denotes the root directory of the reproducibility package.

Output Files

After a successful execution of `RunAll.ps1`, intermediate outputs can be found in `./Output/`, while all reproduced figures and tables can be found in `./Result/`.

Please note that the reproduced figures and tables are not numbered. Figure and table numbering in the manuscript is generated during the LaTeX compilation process and is therefore not reproduced by the scripts.

For convenience, the following sections provide a mapping between the manuscript figure/table numbers and the corresponding output files.

Reference Results, Reproduced Results, Result Mapping

The files listed below are available in both `./Result` (reproduced outputs) and `./reference` (author-provided reference outputs). Replicators may compare the corresponding files to

verify successful reproduction.

Please take into account the discrepancies described in the [Expected Discrepancies](#) section when comparing reproduced outputs against the reference outputs.

The script `RELEASE.FigureTable.py` uses `#region` and `#endregion` markers extensively. Replicators may use code-folding features available in editors such as VS Code to navigate the script more efficiently.

Numeration in Manuscript	Line in <code>RELEASE.FigureTable.py</code>	File Name
Figure 1	Line 190	<code>Fig.nan_describe.jpeg</code>
Figure 2	Line 235	<code>Fig.bound_describe.jpeg</code>
Figure 3	Line 301	<code>Fig.nu_describe.jpeg</code>
Table 1	Line 370	<code>Table.CRPS_Skill_Mainpaper.csv</code>
Table A	Line 412	<code>Table.CRPS_Skill_Appendix.csv</code>
Table 2	Line 453	<code>Table.Rank.csv</code>
Figure 4	Line 498	<code>Fig.Skill.jpeg</code>
Figure 5	Line 558	<code>Fig.Reliability.jpeg</code>
Figure 6	Line 731	<code>Fig.case_study.jpeg</code>
Figure 7	Line 894	<code>Fig.Sensitivity.jpeg</code>
Figure A	Line 1139	<code>Fig.DM_Metric_Heatmap.png</code>

Codebook

We use variable names that are consistent with the notation adopted in the manuscript.

Please note that not every variable appearing in the manuscript is explicitly implemented in the software. Only variables required for model implementation and computation are included in the source code.